

Owen Skriloff

CONTACT INFORMATION	Tufts University Department of Mathematics 177 College Ave, Medford, MA 02155	https://oskril01.github.io owenskriloff@me.com
RESEARCH INTERESTS	I want to understand how prior scientific knowledge shapes machine learning architecture choices. My approach uses geometric, statistical, and algebraic tools to probe the latent representations of models. By studying how constraints influence both the training process and the resulting latent spaces, I aim to develop systematic methods for designing intelligent systems that fully leverage existing scientific knowledge. I am particularly interested in the application of physics-informed machine learning for chemical and biological problems and the application of tools from statistical physics to data science.	
EDUCATION	MS, Tufts University School of Arts and Sciences <i>Mathematics</i> GPA: 4.0, Concentration in Foundations of Machine Learning	(Expected) May 2026
	BS, Tufts University School of Engineering <i>Chemical and Biological Engineering and Mathematics</i> GPA: 3.9, Highest Thesis Honors, Summa Cum Laude	May 2025
AWARDS	Benjamin G Brown Scholarship (<i>Research Excellence</i>) Christos Georgakis Award (<i>Best Thesis</i>) Chemical Engineering Summer Research Grant Regeneron Science Talent Search Top Scholar National Science and Humanities Symposium: 3rd Place Physical Sciences	2025 2025 2024 2020 2020
PUBLICATIONS	[1] Skriloff, O., D. M. Stoukides, and E. S. Tzanakakis. 2025. Augmentation Models of Stem Cell Culture Data for the Application of Machine Learning. <i>Biotechnology and Bioengineering</i> 0: 1-11. https://doi.org/10.1002/bit.70094 .	
CONFERENCES	[1] O. Skriloff, D. Stoukides, E. Tzanakakis. Data Augmentation Models for Machine Learning Applications in Stem Cell Bioprocessing. <i>AIChE Annual Meeting</i> , 2025. Speaker . [2] O. Skriloff, D. Stoukides, E. Tzanakakis. Data augmentation models for stem cell bioprocessing. <i>Cell Culture Engineering XIX (ECI)</i> , 2025. Poster .	
EMPLOYMENT	Tzanakakis Research Group <i>Researcher</i> • Engineered and trained three physics-informed ML models of stem cell growth dynamics. • Built novel data pipeline with Monte Carlo simulation and feature importance to quantify and visualize representational alignment between models and known biological behavior. • Determined optimal model class and design procedure for stem cell manufacturing.	Medford, MA Jan 2023 - May 2025
	Itaconix Corporation <i>Summer Associate Engineer</i> • Commissioned, purchased parts for, and assembled a 100L SS reactor with vacuum pump. • Scaled an esterification process from 250mL to the first pilot production at 50L. • Developed HPLC and GPC methods for 200+ samples and supported GPC installation.	Stratham, NH Jun 2023 - Aug 2023
	Araca Inc. <i>Laboratory Technician</i> • Conducted 25+ chemical mechanical planarization (CMP) and post-CMP cleaning experiments in a Class 100 cleanroom to optimize CMP conditions for semiconductor manufacturing. • Measured 250+ wafers, mixed slurries, programmed polishers, and tested new CMP products. • Analyzed data in MATLAB, Excel, and LabVIEW to prepare 25+ presentations visualizing experimental results to inform client process improvements.	Tucson, AZ Sep 2020 - Dec 2020

Temple University Smart Biomaterials Lab

Researcher

Philadelphia, PA

Mar 2019 - Aug 2020

- Developed and performed methodology to quantify a relationship between stress and mineral deposits generated by electric charges from a piezoelectric composite using simulated body fluid, incubator, and actuator.
- Authored research report: An *in-vitro* evaluation of the relationship between stress and mineralization through the use of a piezoelectric barium titanate composite.

SELECTED PROJECTS

Loss-landscape geometry and generalization across MLP depths

Final Project: Iterative Methods in ML

May 2025

- Investigated how network depth affects generalization by analyzing typical loss landscape curvature in both Euclidean and Fisher-Rao geometry with random training restarts.
- Improved alignment of training with validation loss by 15% using geometric regularization.
- Discovered empirical broken scaling law ($R^2 = 0.93$) between loss landscape curvature and validation loss, with scaling break (critical point) at optimal depth.

Industrial amikacin production plant: process design and economic analysis

CHBE Capstone Project: With David Gillingham, Ivy Le, Youngseo Yi

May 2025

- Designed and simulated industrial scale amikacin synthesis process including bioprocessing and chemical reaction steps using SuperPro, Excel, and Python.
- Performed Class IV economic analysis of profitability (labor, materials, and equipment costs).

Energy and Entropy Measures for Reaction Diffusion PDE Analysis

Final Project: Numerical PDEs

Dec 2024

- Designed, proved stability and convergence, and implemented efficient finite element numerical solver in Python for reaction diffusion systems.
- Analyzed efficacy of entropy and energy measures for detecting emergent dynamic behavior.

APPLICABLE COURSES

Mathematics

Numerical PDEs, Iterative Methods in Machine Learning, Statistics, High Dimensional Probability, Mathematical Aspects of Data, Mathematical Modeling, Analysis, Algebra, Differential Geometry.

Chemistry and Chemical Engineering

Process Dynamics & Control, Transport Phenomena, Thermodynamics, Chemical Physics, Intro to Modern Physics, Polymerization, Materials Science, Numerical Methods, Organic Chemistry.

SKILLS

Computer Languages

Python, MATLAB, LaTeX, HTML.

Scientific Computing

Neural networks (PyTorch, TensorFlow), data science (scikit-learn, NumPy, pandas), data visualization (seaborn, matplotlib), version control (Git, Unix, Bash), object-oriented programming, Monte Carlo simulation, optimization, finite element solvers.

Engineering

Reaction design and scale-up, ChemStation, HPLC and GPC, chemical mechanical planarization, reaction-kinetic modeling, process control modeling, SuperPro Designer.

PERSONAL INTERESTS

Outside of academics, I'm an avid reader and artist. I love to draw, paint, play saxophone, and spend time traveling and in museums. I also enjoy skiing, running, hiking, and doing flips.